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Prof. Alán Aspuru-Guzik
Editor-in-Chief, *Digital Discovery*
Royal Society of Chemistry

Dear Professor Aspuru-Guzik,

On behalf of all authors, we submit the manuscript entitled “An open-source, 3D-printed auger powder doser designed with generative artificial intelligence,” by Sam Charles, Will Mulberry, Luke Winters, Carl Robison, Gage Erickson, and Sterling G. Baird, for consideration in *Digital Discovery* as a Full paper describing open-source scientific hardware, and ask that it be considered for the journal’s New Hardware Developments collection.

In this work, we present a single-channel powder doser that a laboratory can print, build, and modify for under \$200, plus an analytical balance. A \$6 microcontroller turns, tilts, and taps a printed auger (a tube with a spiral inside) until the balance reads the requested mass. We tested 13 powders, seven food-safe and six relevant to alloy discovery. With one fixed set of settings, ten moved 10–231 mg per auger revolution, and three moved no measurable amount. With the balance in the loop, 24 of 39 doses of 1 g finished within $\pm 5\%$ of the target, including every dose on the seven fastest-flowing powders dosed; the nine doses on three alloy-relevant powders were all within 1%. Smaller doses were less reliable (14 of 30 at 50 mg and 18 of 30 at 200 mg met their limits), and a typical dose took minutes rather than seconds. The Supplementary Information (SI) accounts for every dose attempt, failures included.

The second contribution is a documented case study of AI-assisted hardware design. AI coding agents and a chat-driven CAD tool first modelled the mechanical parts, while the team made every design decision and reviewed, printed, and tested every part. Asking for the whole assembly at once failed repeatedly; asking for one part at a time, against dimensioned drawings, worked better. We judged every AI route to be slower than an experienced engineer using conventional CAD, and its geometry poorer. The team later redrew the parts in Fusion 360, and the tested doser uses some of them, including the auger and its drive gear. What the AI route gave us instead was a public, written record of every design decision.

We believe the paper suits *Digital Discovery* and this collection for two reasons. First, it brings open-source hardware to a step that most self-driving laboratories either avoid or leave to people, in part because commercial dosing systems cost tens to hundreds of thousands of dollars. Built for alloy discovery by metal 3D printing, the doser is already in use beyond the tests reported, for example weighing AlSi10Mg for a charge that our ultrasonic atomizer turned into new powder. Second, its design record, including a 128-entry design log, shows where generative AI helped and where it did not, although it comes from one team and one project rather than a controlled comparison.

Referees can check everything now, without requesting access. The SI gives the bill of materials and a construction guide, and the repository at <https://github.com/vertical-cloud-lab/powder-doser> holds the design files, firmware, raw test records, analysis scripts, design log, and the dated prompts and responses exchanged with the AI tools. A tagged release is archived on Zenodo ([TODO: Zenodo DOI]).

The manuscript and SI have been publicly visible in this repository while we drafted them; [TODO: preprint status, e.g. “a preprint is on ChemRxiv (DOI)” or “no preprint has been posted”].

As RSC policy requires, we declare that generative AI was both a subject of this study and an assistant in it. AI coding agents running Anthropic’s Claude models (GitHub Copilot and Claude Code) wrote CAD code, firmware, and analysis and figure scripts, and drafted and revised the manuscript, the SI, and this letter. Edison Scientific searched the literature, compiled references, and ran mock reviews, and CADSmith and Zoo Design Studio served as text-to-CAD tools. The authors reviewed and edited all output and take full responsibility for the content; the Experimental section, the Acknowledgements, and the SI give the tools, models, and dates.

We certify that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. It has been approved by all authors and by any relevant authorities at the locations where the work was performed. The authors declare no conflicts of interest. Sam Charles and Sterling G. Baird are co-corresponding authors.

We look forward to a review of this paper. Please advise if you require anything to proceed.

Sincerely,

[TODO: signature]

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